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Beyond Tools, Toward Power Structures: A Critical Review of AI in Primary Education

Entre las herramientas y las estructuras de poder: una revisión crítica de la inteligencia artificial en la educación primaria

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DOI: 10.5944/reec.48.2025.45126

**Recibido: 10 de abril de 2025
Aceptado: 28 de julio de 2025**

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Abstract

This article critically review how academic literature on artificial intelligence (AI) in primary education addresses the multiple dimensions of its impact on digital literacy moving beyond an instrumental understanding of AI to examine how it also functions as a power structure. 23 primary studies were qualitatively analysed through a seven-dimensional analytical framework: instrumental, epistemological, ethical, social, political, commercial, and ideological. 4 systematic reviews were consulted for context but not coded. Coverage is uneven: instrumental, ethical, epistemological and social dimensions appear most often, and commercial and ideological aspects remain marginal. Correlation analysis shows epistemology at the hub of current debate, yet reveals sharp disconnections—particularly technical vs. governance and market vs. social—that fragment understanding of AI's educational role. These patterns confirm that AI is still framed chiefly as a technical fix with moral overtones, rather than as an entangled power structure shaping classrooms, policies and markets. A multidimensional lens is therefore essential. Only through integrated, critically informed collaboration among educators, families, researchers, industry and policy-makers can AI advance educational justice and democratic participation, instead of deepening existing inequities.

Keywords: Artificial Intelligence in Education; Digital Literacy; Critical Framework; Primary Education

Resumen

Este artículo revisa críticamente cómo la literatura académica sobre inteligencia artificial (IA) en educación primaria aborda las múltiples dimensiones de su impacto en la alfabetización digital, y supera una comprensión meramente instrumental de la IA para examinar también su funcionamiento como estructura de poder. Mediante un análisis cualitativo de 23 estudios primarios (se consultaron 4 revisiones sistemáticas solo para contextualizar, pero no se codificaron), se aplica un marco analítico de siete dimensiones: instrumental, epistemológica, ética, social, política, comercial e ideológica. La cobertura es desigual: las dimensiones instrumental, ética, epistemológica y social son las más tratadas, mientras que los aspectos comercial e ideológico siguen siendo marginales. El análisis de correlaciones sitúa la epistemología en el centro del debate, pero revela marcadas desconexiones—en especial entre lo técnico y la gobernanza, y entre lo mercantil y lo social—que fragmentan la comprensión del papel educativo de la IA. Estos patrones confirman que la IA continúa enmarcándose sobre todo como una solución técnica con matices morales, más que como una estructura de poder entrelazada que moldea aulas, políticas y mercados. De ahí la necesidad de una mirada multidimensional. Solo una colaboración integrada y críticamente informada entre docentes, familias, investigadores, industria y responsables políticos permitirá que la IA favorezca la justicia educativa y la participación democrática en lugar de profundizar las desigualdades existentes.

Palabras clave: Inteligencia artificial en educación; Alfabetización digital; Marco analítico crítico; Educación primaria

1. Introduction

The acceleration of Artificial Intelligence (AI) and digital transformation is reshaping global education, largely driven by capitalist expansion, digital governance, and emerging policy frameworks (Macgilchrist, 2021; Selwyn, 2024, p. 202; Williamson, 2017). While AI is often discussed in terms of its technical affordances and potential for enhancing digital literacy (Holmes *et al.*, 2022), critical perspectives emphasize its broader implications, including epistemological, ethical, social, political, ideological, and commercial dimensions (Jandrić *et al.*, 2018; Knox, 2020; Williamson & Hogan, 2020). However, these perspectives are often addressed in fragmented ways, limiting a comprehensive understanding of AI's role in education and its integration into policy and practice (Facer & Selwyn, 2021; Macgilchrist *et al.*, 2022).

This study builds upon the framework proposed by Castañeda *et al.* (in press) to analyse how AI is conceptualized in the existing literature on AI in primary education. Using this framework, we explore whether and how different dimensions are considered in academic research, identifying patterns, gaps, and dominant narratives that shape the discourse on AI (Bond *et al.*, 2024; Zawacki-Richter *et al.*, 2019). Rather than focusing solely on technical applications, we examine the intersections between AI and broader educational, social, and economic structures, offering a nuanced perspective on its role in digital transformation (Ames, 2018).

By positioning this analysis within the broader discussions on the global EdTech industry, digital governance, and the privatization of education (Williamson, 2019), this study aims to inform future research, policy discussions, and theoretical developments, fostering a more interconnected and critical understanding of AI's role in shaping contemporary education systems.

2. Dimensions for Analysing AI's impact on Education

AI's growing presence in K-12 education requires an expanded perspective beyond simple considerations, urging a deeper reflection on its instrumental/technical, epistemological, ethical, social, political, commercial, and ideological implications. Thus, the framework we are going to use for doing this analysis (Castañeda, *et al.* In Press), included in Figure 1, is built upon seven interconnected dimensions that provide a holistic approach to understanding AI role in education.

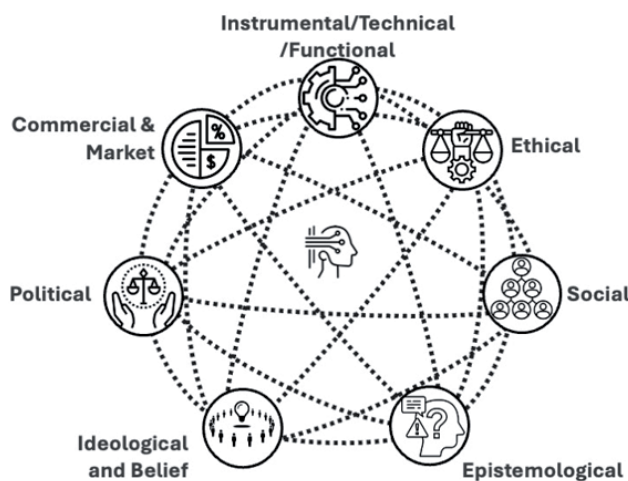


Figure 1. Dimensions for Analysing AI's impact on Education. Castañeda *et al.* (in press)

The instrumental, technical, and functional dimension considers AI's ability to enhance digital literacy by personalizing learning, automating tasks, and providing real-time feedback. Drawing on post-digital and sociomaterial scholarship, we treat the "technical" as a sociomaterial entanglement—an effect that emerges through the relations among AI systems, pedagogical practices, and institutional routines within the classroom. Challenges such as accessibility, infrastructural limitations, and the transparency of algorithmic decision-making must be addressed. The potential for over-reliance on AI, where users unquestioningly trust its outputs without understanding its limitations, is an issue that requires careful consideration.

The epistemological dimension questions how AI reshapes knowledge, learning, and truth. Unlike human cognition, AI generates outputs without understanding, raising fundamental questions about what constitutes knowledge and how learning should be approached in an era where machines can analyse data and produce information. The role of education (and, hence, teachers) in this context must be to equip students with the ability to critically engage with AI-generated knowledge rather than passively accepting it.

Ethical considerations form another crucial dimension, as AI's integration into education raises concerns about transparency, bias, privacy, and student agency. The framework highlights the necessity of clear accountability mechanisms to prevent AI from undermining human judgment, reinforcing biases, or infringing on privacy. Ethical implementation demands that it supports rather than diminishes autonomy in learning. This ethical discussion explicitly incorporates an axiological perspective, foregrounding the design values and worldviews inscribed in AI systems and examining how these values align—or collide—with educational purposes.

The social dimension focuses on AI's influence on relationships within educational settings and surrounding communities. From an anthropological perspective, this dimension also interrogates how daily encounters with AI mediate children's identity formation, cultural participation and meaning making within families, peer groups and school communities. AI tools have the potential to support learning, yet they also risk eroding interpersonal relationships, reducing opportunities for cultural engagement, and deepening educational inequalities by privileging those with greater access to technological resources.

On a political level, the framework examines how AI is not merely a neutral tool but an actor that affects governance, policymaking, and decision-making within educational institutions. AI can be leveraged to optimize resources and curriculum planning, but it also has the potential to concentrate power in the hands of governments or private corporations. Transparency in AI-driven decisions, as well as efforts to educate students about AI's role in shaping democratic participation, are essential to prevent it from becoming an opaque force that dictates educational policies without public scrutiny.

The commercial and market dimension addresses the economic realities of AI-driven education, questioning its sustainability and equity. The growing presence of AI in education is closely tied to private corporations that develop and control these technologies. This raises concerns about accessibility, as wealthier institutions may benefit disproportionately from AI's capabilities while underfunded schools struggle to implement them. The increasing privatization of AI in education risks transforming it into a commodity rather than a public good, necessitating policies that safeguard educational equity and prevent commercial interests from taking precedence over pedagogical objectives.

Finally, the ideological and belief dimension explores the cultural narratives surrounding AI and its influence on education. AI is often framed within either a technoutopian vision, seen as a solution to all educational challenges, or a technophobic perspective, feared as a threat to traditional teaching. The framework challenges both extremes, advocating instead for a critical, balanced approach that acknowledges AI's potential while recognizing its risks. By fostering digital literacy that includes awareness of AI's biases, cultural assumptions, and political implications, educators can ensure that students engage with AI not as passive consumers but as informed participants in a technologically evolving world.

We do not treat pedagogy or education as an additional, isolated dimension; rather, education constitutes the arena where the seven dimensions intersect and materialise, for example, in classroom practice, making visible how sociotechnical arrangements reshape planning, instruction and assessment.

To support the practical application of this framework, guiding questions accompany each of these seven dimensions. These questions serve as a reflective tool for educators, policymakers, and stakeholders to critically assess how AI is being integrated into digital literacy education. They encourage considerations of AI's benefits and challenges, prompting discussions that extend beyond technical efficiency to address broader educational, ethical, and societal concerns.

By providing this structured yet flexible approach, the framework ensures that AI is not simply adopted as a technological solution but is scrutinized in a way that aligns with educational values, human agency, and social responsibility.

3. Method

This study aims to critically examine how the existing literature on artificial intelligence (AI) in primary education addresses the multiple dimensions of AI's impact on digital literacy. Rather than limiting the discussion to the technical affordances of AI in schools, the objective is to provide a comprehensive analysis that explores epistemological, ethical, social, political, commercial and ideological dimensions. By doing so, the study seeks to identify patterns, gaps, and prevailing narratives in the academic discourse, offering insights into the broader implications of AI integration in primary education.

To achieve this objective, we conducted a structured literature review based on a systematic search across major academic databases (Zawacki-Richter *et al.*, 2020). The search process involved developing a comprehensive search string structured around four key components: primary education, artificial intelligence, digital literacy, and academic reports. Each component was expanded with synonyms and related terms to ensure inclusivity. The AI-related terms were adapted from the metareview by Bond *et al.* (2024), refining the terminology to align with established academic discourse. Additionally, AI-assisted tools (*Prompt-perfect* <https://promptperfect.jina.ai/>) were used to enhance precision, with final adjustments made through human supervision to maintain relevance. The complete search string-building process is provided in Figure 2.

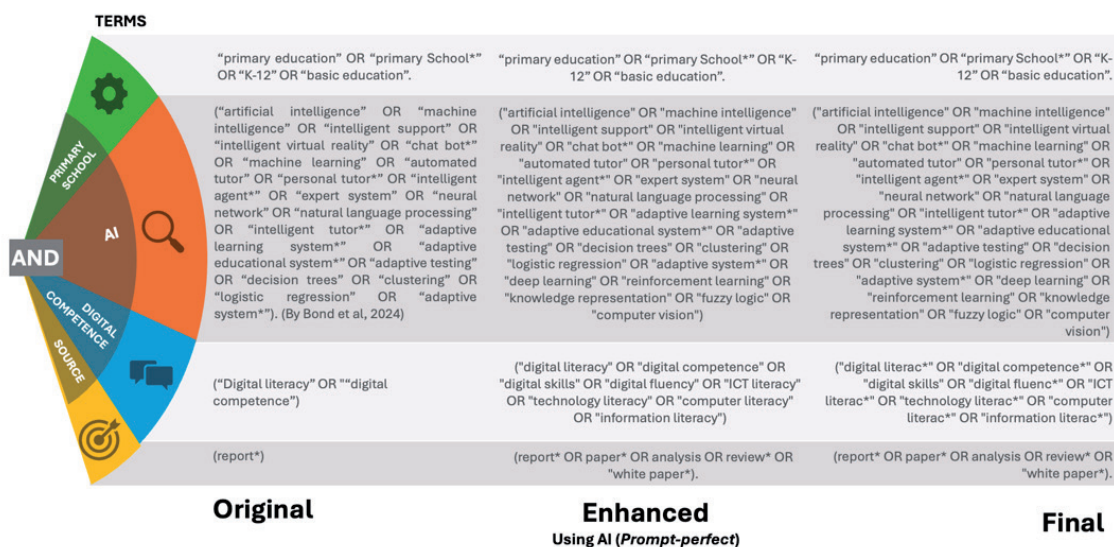


Figure 2. Search-string building process

The search was conducted on the Web of Science (WOS), Scopus, and Google Scholar. In WOS, 23 articles were identified, while Scopus initially retrieved 48, which was later refined to 29 after excluding conference papers. Google Scholar yielded over 18,000 results, which were narrowed down to 260 using the *allintitle* function. Further filtering to include only publications from 2022 to 2025 reduced this number to 147.

A time limitation was applied, restricting the review to studies published from 2022 onwards to capture the most recent developments in AI adoption in primary education. This period coincides with the widespread release of generative AI technologies, particularly after OpenAI's ChatGPT became publicly available in late 2022 (Mintz *et al.*, 2023). Given the acceleration of AI-driven discussions in education following this milestone, this time frame ensures that the analysed literature reflects the current landscape of AI integration in primary education. Earlier studies, while informative, often predate the large-scale implementation of AI tools in schools and may not fully capture recent trends and challenges.

To ensure the relevance and quality of the dataset, strict inclusion and exclusion criteria were applied. Studies had to be peer-reviewed journal articles or book chapters, explicitly reference AI in primary education, and include a discussion on digital literacy. Excluded materials comprised conference proceedings, non-peer-reviewed documents, studies unrelated to primary education, and inaccessible sources. Additionally, even if we try to cover a more international perspective (Macgilchrist *et al.*, 2022), documents in languages other than English, Spanish, Italian, Portuguese, or German were omitted. After filtering and deduplication, the final dataset consisted of 27 studies, with six from WOS and Scopus and twenty-two from Google Scholar. Four of these items were themselves systematic literature reviews. To avoid 'reviewing reviews', we retained them only for mapping the field and excluded them from the qualitative coding and correlational analysis. The analytic corpus therefore comprises 23 primary studies. Even though they were not part of the formal analysis, some literature reviews were revisited to support reflective observations in the conclusion.

Each of the 23 primary studies was analysed using a directed (deductive) content analysis approach, applying the seven-dimensional framework to identify explicit and

implicit references to AI's impact in primary education. We have used a double cycle coding strategy: a descriptive one, focused on recognising direct mentions of each dimension, detecting related discussions even when not explicitly framed within the framework, and a second one, thematic, extracting qualitative excerpts illustrating key themes (Saldaña, 2015). The analysis explored patterns across studies, considering differences in publication year, study type, and depth of engagement with each dimension. Special attention was given to underexplored areas, highlighting gaps in the literature and assessing how well current research addresses AI's broader educational implications.

Although analyses such as co-occurrence matrices or distribution comparisons could highlight patterns in how AI analysis levels relate to each other, they do not align with our primary objective. Given the limited sample size, these methods may suggest statistical associations, but they do not capture the complex, multifaceted understanding we seek regarding AI's role in education. Rather than focusing on numerical relationships, we prioritize a deeper exploration within each level of analysis, as this approach allows us to critically examine how AI is conceptualized and the diverse perspectives that shape its integration in educational research.

4. Results

4.1. The Analysed Sample

The analytic sample on which the results are based consists of 23 primary studies (19 journal papers, two research articles and two book chapters). The four systematic reviews identified were used only for contextual mapping and are not included in the analyses reported below with the other papers. The distribution by publication year reveals a growing interest in the topic, with most studies published recently (16 in 2024, 9 in 2023, and only 2 before 2023). Regarding the nature of the studies, the majority are empirical investigations (16 articles), while systematic reviews (4 articles) and conceptual or theoretical contributions (including essays, editorials, and historical reviews) represent a smaller portion of the sample. This diversity of study types suggests a balance between empirical evidence and theoretical discussions on the role of AI in education.

In terms of AI analysis levels, the instrumental/technical, epistemological and political dimensions are the most frequently addressed, with 12 articles explicitly including them, while the ethical dimension is even more prevalent, appearing in 14 studies. The social dimensions receive moderate attention, included in 10 studies, but often appear in an implicit manner in additional papers. Meanwhile, the commercial/market and ideological/belief dimensions are the least explored, with only 5 and 7 articles explicitly discussing them, respectively, and a significant number of studies omitting these aspects entirely. These results highlight a tendency in the literature to focus on functional and ethical considerations of AI in education, while aspects related to governance, market dynamics, and ideological implications remain less developed. A visual summary of these findings is presented in Figure 3, where the distribution of AI analysis levels across the sample can be observed in detail.

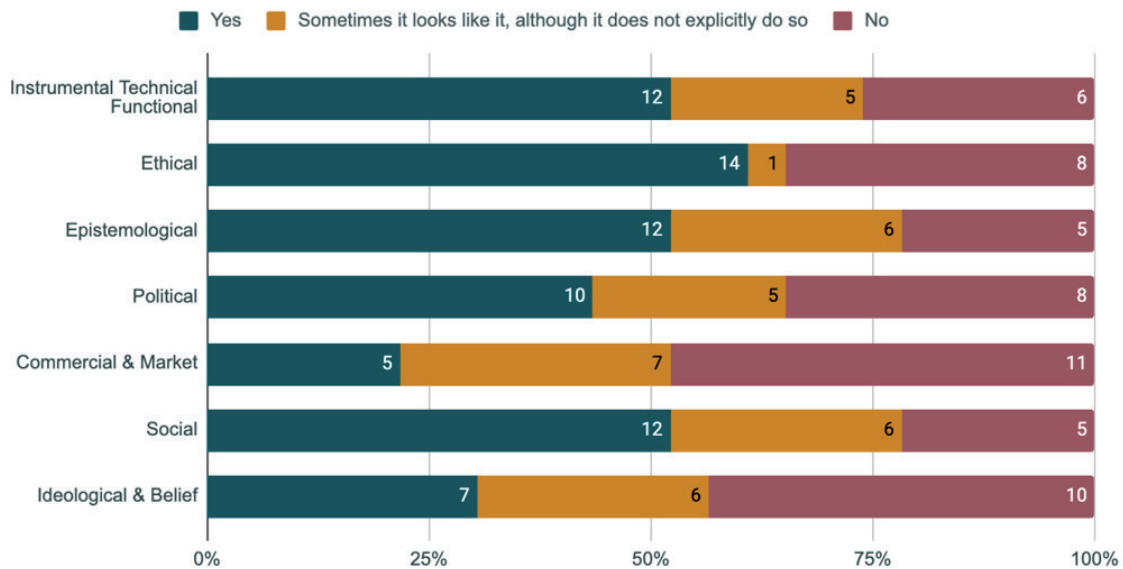


Figure 3. AI in Primary School literature Review. Thematic analysis

A correlation analysis provides insights into the relationships between the different AI analysis levels explored in the literature. To examine these connections, we first recoded the articles based on the presence of each AI analysis level. Articles that included an explicit mention of a given level were assigned a 1, those that contained related but non-explicit content were coded as 0.5, and those that did not mention the level were assigned a 0. Using this recoded dataset, we then calculated Pearson correlation coefficients among the different dimensions using Excel's Analysis ToolPak, generating a correlation matrix that quantifies the degree to which each level of analysis is associated with the others. Pearson's correlation measures the linear relationship between variables, with values ranging from -1 (strong negative correlation) to 1 (strong positive correlation), while values close to 0 indicate no meaningful relationship. The results of this analysis are presented in Table 1, which displays the correlation matrix.

Table 1.

AI correlational Analysis of the Levels of analysis appeared in the articles

	Technical	Ethical	Epistemological	Political	Commercial	Social	Ideological
Technical	1,00	0,17	-0,03	-0,18	-0,12	0,52	-0,05
Ethical	0,17	1,00	0,43	0,13	0,29	0,00	0,52
Epistemological	-0,03	0,43	1,00	0,39	0,71	0,62	0,54
Political	-0,18	0,13	0,39	1,00	0,16	0,16	0,22
Commercial	-0,12	0,29	0,71	0,16	1,00	-0,07	0,03
Social	0,52	0,00	0,62	0,16	-0,07	1,00	0,24
Ideological	-0,05	0,52	0,54	0,22	0,03	0,24	1,00

The findings suggest that epistemological considerations are central to the literature on AI in education, as they show the strongest associations with multiple other levels. The correlation between epistemological and commercial aspects (0.71) suggests that studies

focusing on AI's role in knowledge construction often also address its economic implications. Similarly, the epistemological-social correlation (0.62) indicates that discussions on AI's epistemic impact frequently extend to its broader societal effects. The ethical and ideological dimensions (0.52) are also moderately correlated, reflecting that ethical considerations in AI education are often connected to broader ideological debates.

On the other hand, some relationships appear much weaker, highlighting gaps in the literature. The technical and political dimensions (-0.18) show a slight negative correlation, suggesting that studies on AI governance do not typically engage deeply with its technical foundations. Likewise, the ethical and social dimensions (0.00) appear largely independent, indicating that ethical debates around AI in education do not necessarily align with broader societal discussions. Furthermore, the correlation between technical and commercial dimensions (-0.12) suggests that AI's functional aspects are rarely examined alongside its market implications. The near-zero or negative correlations in these areas underscore a fragmented understanding of how different layers of AI's influence intersect in educational research.

4.2. Technical dimension

Out of the 23 studies analysed, 12 explicitly address the instrumental, technical, and functional dimension (Arnone *et al.*, 2023; Chen, 2024; Cheng & Wang, 2023; Chiu, Ahmad, *et al.*, 2024; Duisenova & Zhorabekova, 2024; Hong & Kim, 2024; Mintz *et al.*, 2023; Pörn *et al.*, 2024; Webb *et al.*, 2023; Woodruff *et al.*, 2023; Yeter *et al.*, 2024; Zhao, 2022). Five studies engage with instrumental, technical, and functional aspects in an implicit manner (Chiu, Falloon, *et al.*, 2024; Li, 2024; Omae *et al.*, 2023; Velandar *et al.*, 2024; Yue *et al.*, 2024), while the remaining six do not address these themes at all.

Since the advent of generative AI, the technical issues of AI have been the most recognised. Possibilities such as the diversification of educational materials by allowing for adaptation across different languages, proficiency levels, and formats are examples of the highlighted functional potential of AI in reshaping instructional and learning practices (Chen, 2024). In the case of generative AI, through its natural language processing functions, it facilitates interactive educational experiences and supports access in multiple languages, thereby enhancing inclusivity and fostering the development of linguistic skills (Duisenova & Zhorabekova, 2024).

In this context, Webb *et al.* (2023) recall the classification of the Luckin's team for AI in education (AIED) that defined three types of systems: 1) Learner-facing AIED systems, 2) Educator-facing AIED systems, and 3) AIED systems for institutional support. This typology helps to understand the varied educational stakeholders involved and the complexity of the techno-educational ecosystem.

A recurring theme in the literature is the dual role of AI (Yeter *et al.*, 2024; Duisenova & Zhorabekova, 2024): as a tool for enhancing AI literacy among students, and as a means of optimizing educational processes. In primary (and secondary) education, interactive platforms such as teachable machines like LearningML, block-based programming environments like Scratch, and creative tasks are increasingly used to scaffold students' understanding of AI concepts, offering real-time feedback and promoting engagement. Simultaneously, AI applications like automated grading and intelligent tutoring systems are examples of the potential of the technology to streamline routine educational tasks, reducing teacher workload (Arnone *et al.*, 2023) and allowing educators to focus on more complex pedagogical goals (Duisenova & Zhorabekova, 2024). Additionally, tools

for instructional design and assessment support (Pörn *et al.*, 2024), as well as collaborative robots (“cobots”) for instructional and administrative tasks (Woodruff *et al.*, 2023), theoretically expand the functional potential of AI.

The literature also highlights several learner-centered innovations, including personalized learning, real-time feedback, adaptive tutoring and learning systems, virtual writing assistants, etc. (Mintz *et al.*, 2023). However, despite the promise of AI to transform educational experiences, instrumental limitations remain a critical challenge which deeply influence the integration of AI in early childhood and elementary education (Arnone *et al.*, 2023). Issues such as software design, hardware accessibility, and ease of use - particularly in relation to teachers’ digital skills- can hinder effective implementation.

Furthermore, a gap persists between the potential envisioned by researchers and the current technical capabilities of AI systems and developers. More critically, doubts persist about the broader value of AI in K–12 contexts, where large-scale effectiveness, pedagogical integration, and the evolving role of teachers remain underexplored and unclear (Chiu, Falloon, *et al.*, 2024).

Regarding teacher preparation, Chiu *et al.* (2024) establish foundational AI knowledge as a key dimension in evaluating teacher competence in the context of AI integration. The Teacher AI Competence Self-Efficacy Scale - TAICS (Chiu, Moorhouse, *et al.*, 2024) defines that foundational knowledge as recognizing whether a tool is AI-based, creating content using AI, and explaining core concepts such as artificial intelligence, machine learning, deep learning, generative AI, cloud computing, and big data. Hong & Kim (2024) propose to integrate other key functions such as data analysis through sensor modules, autonomous decision-making based on AI-processed information, and adaptive learning capabilities. That foundational knowledge, combined with the multidisciplinary nature of AI requests, demands an integrative approach that should boost students’ AI literacy in terms of helping them to develop a broader academic perspective and situate AI within a wider learning context (Zhao, 2022), avoiding their over-reliance on AI.

The instrumental, technical and functional aspects of AI both enhance and condition the integration in education: both opportunities and challenges require a focus on AI literacy and understanding of AI-based systems. Educational stakeholders and the entire educational community have a responsibility to respond to this immediate need, as well as to prepare also for technology-mediated decision-making.

4.3. Epistemological dimension

From the 23 studies analysed 12 explicitly examine epistemological aspects, such as AI’s role in knowledge creation and validation (Chen, 2024; Duisenova & Zhorabekova, 2024; Hong & Kim, 2024; Kong & Yang, 2024; Lucas *et al.*, 2024; Mintz *et al.*, 2023; Mora Aristega *et al.*, 2023; Pörn *et al.*, 2024; Skaar, 2024; Woodruff *et al.*, 2023; Yue *et al.*, 2024; Zhao, 2022). 6 refer to epistemological concerns without substantial analysis (Cheng & Wang, 2023; Chiu, Falloon, *et al.*, 2024; Du & Lv, 2024; Li, 2024; Webb *et al.*, 2023; Yeter *et al.*, 2024). The remaining 5 do not discuss epistemological aspects.

The integration of AI in education presents significant epistemological challenges, reshaping knowledge creation, validation, and transmission. AI’s ability to generate and organize vast amounts of information challenges traditional understandings of knowledge, which have been rooted in human cognition, intentionality, and social collaboration. AI-generated knowledge, lacking human intentionality and operating probabilistically, raises concerns about its epistemic status (Skaar, 2024). Determining whether

AI-generated outputs qualify as knowledge or merely statistical patterns requires new validation frameworks (Lucas *et al.*, 2024).

As AI contributes to knowledge production, concerns regarding reliability, fairness, and bias become critical. AI operates through complex algorithms, often lacking transparency, making it difficult to assess its trustworthiness (Pörn *et al.*, 2024). Teachers express skepticism toward AI-generated recommendations, particularly when they conflict with human expertise (Lucas *et al.*, 2024).

AI transforms learning models by introducing personalized and adaptive experiences. Traditionally, education followed sequential, cumulative knowledge-building processes, whereas AI facilitates non-linear learning pathways, enabling individualized progression and real-time feedback (Duisenova & Zhorabekova, 2024). This fosters epistemic autonomy but also raises concerns about over-reliance on AI, which may diminish independent critical thinking (Hong & Kim, 2024). AI-based tools personalize learning, moving away from one-size-fits-all models and enhancing student achievement (Woodruff *et al.*, 2023).

The rise of AI necessitates re-evaluating educational priorities. Traditional IT education emphasized digital literacy and technical skills, but as AI becomes pervasive, curricula must prioritize AI literacy, interdisciplinary thinking, and ethical considerations (Zhao, 2022). Pedagogical approaches must shift, as AI alters traditional instruction methods. Teachers must transition from knowledge transmitters to mentors who guide students in AI-enhanced environments (Duisenova & Zhorabekova, 2024). This transformation demands significant pedagogical adjustments, including teacher training and curriculum redesign (Mintz *et al.*, 2023). Professional organizations advocate for AI integration to improve learning outcomes across disciplines (Woodruff *et al.*, 2023).

With AI facilitating knowledge access, teachers must help students critically engage with AI-generated information while ensuring deep intellectual engagement (Pörn *et al.*, 2024). AI should be integrated without diminishing teachers' roles in fostering creativity, ethical reasoning, and critical inquiry (Lucas *et al.*, 2024). AI should complement human creativity, encouraging students to use AI for idea generation rather than accepting AI-generated solutions uncritically (Hong & Kim, 2024). Additionally, AI tools encourage students to develop computational thinking, aligning with 21st-century educational standards (Woodruff *et al.*, 2023).

AI systems, trained on vast datasets, often reflect biases, reinforcing stereotypes and marginalizing perspectives (Chen, 2024). To counteract this, ensuring culturally relevant and inclusive AI-generated content is essential for equitable knowledge representation (Chen, 2024).

Moreover, studying AI in education requires robust theoretical frameworks to avoid superficial understandings of its implications (Skaar, 2024). Viewing AI merely as a technological tool is insufficient; research must incorporate multiple epistemological perspectives to fully grasp how AI reshapes knowledge construction (Skaar, 2024). Combining theoretical insights with practical applications ensures AI enhances both knowledge production and educational practice (Chen, 2024). Balancing theoretical depth with real-world application is crucial, emphasizing that AI education should bridge conceptual understanding with hands-on experiences (Chen, 2024). AI's role in project-based learning allows students to engage with real-life problem-solving, enhancing both conceptual and practical knowledge (Hong & Kim, 2024).

4.4. Ethical dimension

14 studies, from the 23 analyzed, directly discuss ethical aspects of AI, such as fairness, bias, or moral implications (Chen, 2024; Chiu *et al.*, 2024; Djouiba, 2024; Duisenova & Zhorabekova, 2024; Kong & Yang, 2024; Mintz *et al.*, 2023; Mora Aristega *et al.*, 2023; Pörn *et al.*, 2024; Skaar, 2024; Webb *et al.*, 2023; Woodruff *et al.*, 2023; Yeter *et al.*, 2024; Yue *et al.*, 2024; Zhao, 2022). 1 mentions ethical concerns without making them a central focus (Cheng & Wang, 2023) and the remaining 8 do not address ethical aspects. Notably, several studies use the term “ethical” as a catch-all category for any non-technical concerns, often labeling issues as ethical without engaging with ethical reasoning or frameworks.

A key ethical challenge is transparency in AI decision-making. Educators, students, and administrators must understand AI tools’ processes. A lack of clarity fosters skepticism, as teachers may distrust AI-generated evaluations (Yeter *et al.*, 2024; Yue *et al.*, 2024). Ensuring AI systems are explainable and interpretable is essential for trust and preventing misuse (Pörn *et al.*, 2024).

AI can personalize learning but may undermine student autonomy and critical thinking. Automated learning paths and AI-driven assessments risk fostering passive learning (Duisenova & Zhorabekova, 2024; Pörn *et al.*, 2024). Additionally, educators may lose professional autonomy if AI dictates curriculum structures and instructional methods (Pörn *et al.*, 2024). AI should support human decision-making rather than replace it.

AI has the potential to reduce educational disparities but may also exacerbate them. Disparities in technological infrastructure and resource allocation risk deepening educational inequalities (Pörn *et al.*, 2024; Yue *et al.*, 2024). AI implementation should prioritize inclusivity to ensure equitable access (Zhao, 2022). Policies must address affordability, digital literacy, and infrastructure readiness to benefit all students (Yeter *et al.*, 2024).

Bias in AI-driven education tools is a significant ethical concern. AI systems are only as unbiased as their training data and may reinforce disparities by privileging dominant cultural norms, languages, and socioeconomic contexts (Woodruff *et al.*, 2023). Ensuring AI tools reflect diverse perspectives is crucial for fairness and non-discriminatory outcomes. Beyond technical considerations, AI must align with cultural and social contexts for ethical and effective use. AI-powered education tools should be designed with cultural sensitivity, recognizing diverse linguistic and educational traditions (Duisenova & Zhorabekova, 2024). Failure to consider cultural variations may reinforce rather than resolve educational inequities (Chen, 2024). Ethical AI literacy in curricula can equip students to critically engage with AI from a culturally aware perspective (Chen, 2024).

Schools and policymakers should implement governance models emphasizing accountability, transparency, and human-centered AI use (Mora Aristega *et al.*, 2023). Ethical review boards and interdisciplinary collaborations can help shape AI policies aligned with educational values.

The commercialization of AI in education raises concerns about data exploitation and commodification of student information. AI tools designed primarily for profit rather than pedagogy risk prioritizing commercial objectives over educational integrity. Schools must safeguard against predatory data practices to ensure AI enhances learning rather than enabling commercial surveillance (Mintz *et al.*, 2023).

4.5. Social dimension

12 of the analyzed studies engage with social dimensions of AI, such as its societal impact and equity (Chen, 2024; Chiu, Moorhouse, *et al.*, 2024; Du & Lv, 2024; Duisenova & Zhorabekova, 2024; Li, 2024; Mintz *et al.*, 2023; Pörn *et al.*, 2024; Webb *et al.*, 2023; Woodruff *et al.*, 2023; Yeter *et al.*, 2024; Yue *et al.*, 2024; Zhao, 2022). 6 touch on social aspects but do not fully explore them (Chiu *et al.*, 2024; Hong & Kim, 2024; Kong & Yang, 2024; Lucas *et al.*, 2024; Omae *et al.*, 2023; Velandar *et al.*, 2024). The remaining 5 do not mention social aspects.

While AI tools offer convenience, over-reliance on AI may weaken direct student-teacher relationships and reduce meaningful interpersonal interactions in the classroom (Duisenova & Zhorabekova, 2024). Teachers express concerns that AI-driven approaches risk making education more impersonal and mechanized, diminishing the human engagement that fosters trust, empathy, and social development (Kong & Yang, 2024; Pörn *et al.*, 2024).

In AI-integrated learning environments, students' emotional and psychological well-being must also be considered. While efficient, automated feedback and monitoring systems can inadvertently increase stress, lower self-esteem, and foster a sense of surveillance (Chiu, Moorhouse, *et al.*, 2024). This is particularly concerning when students experience diminished autonomy and feel disconnected from their learning environments due to AI's impersonal nature (Pörn *et al.*, 2024). Excessive reliance on AI-generated evaluations may also discourage critical self-reflection and reliance on human mentorship (Chen, 2024).

Schools play an essential role in community cohesion, offering spaces for extracurricular activities, cultural events, and interpersonal skill development. An overemphasis on digital platforms may inadvertently weaken these roles, leading to a decline in local engagement and reducing students' exposure to collaborative social environments (Li, 2024).

Parental and community involvement is crucial for the effective implementation of AI in education. Studies highlight that strong parental engagement positively influences AI adoption, although indirect effects suggest that broader contextual factors shape attitudes toward AI use in classrooms (Li, 2024). However, some parents remain sceptical of AI technologies, limiting their children's exposure outside school settings and highlighting the need to consider household digital literacy and access (Du & Lv, 2024). Ensuring that AI-based learning aligns with community values and expectations requires active stakeholder participation, particularly in curriculum development and decision-making processes (Chiu, Moorhouse, *et al.*, 2024).

One of AI's most promising aspects in education is its potential to address structural inequalities. AI can facilitate inclusive learning environments by offering personalized resources for students with disabilities and translation tools for non-native speakers (Duisenova & Zhorabekova, 2024; Yue *et al.*, 2024). However, these benefits are not universally accessible. Schools in underserved regions often lack the necessary digital infrastructure, creating disparities in AI access and exacerbating the digital divide (Yeter *et al.*, 2024; Yue *et al.*, 2024). Policymakers and educators must ensure equitable distribution of AI resources to prevent the deepening of existing socioeconomic inequalities (Yue *et al.*, 2024).

The role of AI in workforce preparation further underscores its significance in education. AI literacy programs are increasingly seen as essential in preparing students for

future careers, equipping them with the skills necessary for a technology-driven job market (Zhao, 2022). However, workforce preparation should not overshadow broader educational goals such as critical thinking, creativity, and ethical reasoning (Mintz *et al.*, 2023). A balanced approach is necessary to ensure that AI integration enhances students' adaptability without reducing education to purely vocational training.

AI also influences the evolving role of teachers, shifting them from traditional knowledge providers to facilitators of AI-enhanced learning experiences (Duisenova & Zhorabekova, 2023). While this change empowers students with greater autonomy, it also challenges conventional educational hierarchies and teaching methodologies (Pörn *et al.*, 2024). Some educators express concerns about the additional workload imposed by AI technologies, fearing that managing AI-driven systems could contribute to teacher burnout and exacerbate workforce attrition (Woodruff *et al.*, 2023). If not implemented carefully, AI may increase bureaucratic burdens rather than reduce them, increasing educators' stress.

4.6. Political dimension

Of the 23 studies analyzed, 10 explicitly address the political dimension of artificial intelligence (Djouiba, 2024; Hong & Kim, 2024; Li, 2024; Lucas *et al.*, 2024; Mora Aristega *et al.*, 2023; Pörn *et al.*, 2024; Webb *et al.*, 2023; Woodruff *et al.*, 2023; Yeter *et al.*, 2024; Yue *et al.*, 2024). 5 studies engage with political aspects in an indirect or implicit manner (Arnone *et al.*, 2023; Du & Lv, 2024; Duisenova & Zhorabekova, 2024; Kong & Yang, 2024; Mintz *et al.*, 2023). The remaining eight studies do not address themes related to the political dimension of AI.

The rapid and unpredictable evolution of AI technologies presents a significant challenge for educational policymakers, who must design adaptable policies capable of keeping pace with ongoing innovation (Yeter *et al.*, 2024). The articles that explicitly address the political dimension of AI in education focus on two main issues: decision-making participation within educational institutions and encouraging political intervention and participation during the education phase.

Regarding decision-making participation, the associated implications of AI integration—including the need for professional development, increased support, and strong community engagement—suggest the importance of broad-based participation for successful implementation in primary education (Li, 2024).

In terms of the promotion of political intervention and participation during the education phase, AI literacy frameworks emphasize the importance of inclusive and accessible AI education to empower individuals to engage meaningfully in societal decision-making processes influenced by AI technologies (Yue *et al.*, 2024). These frameworks are not developed in isolation—they are closely aligned with national and international policy initiatives, such as the EU's Digital Education Action Plan or the framework of the UNESCO's Beijing Consensus—, ensuring that AI education is implemented broadly and coherently across educational systems (Mora Aristega *et al.*, 2023). Despite the persistence of digital divides, government-led initiatives—such as Argentina's *Aprender Conectados* and Uruguay's *Plan Ceibal*—demonstrate proactive efforts to integrate AI and digital literacy into national education systems (Yeter *et al.*, 2024). By grounding AI literacy in public policy, these frameworks aim to equip learners not only with technical knowledge, but also with the civic understanding needed to navigate and shape an AI-driven society.

After all, AI convergence education and AI literacy aims to empower students with the skills needed to address real-world challenges and for decision-making within the context of an AI-driven society (Hong & Kim, 2024); policies should address it, promoting a fairer allocation of resources across educational levels and prioritising schools in rural or remote regions, where limited access and digital infrastructure continues to pose significant challenges (Woodruff *et al.*, 2023).

4.7. Commercial and Market Dimension

As we previously stated, from the 23 studies analysed, five directly discuss these aspects, framing AI as a commodity or economic driver (Hong & Kim, 2024; Mintz *et al.*, 2023; Mora Aristega *et al.*, 2023; Skaar, 2024; Zhao, 2022). Seven mention costs, AI access, or industry ties without a market focus (Du & Lv, 2024; Duisenova & Zhorabekova, 2024; Lucas *et al.*, 2024; Pörn *et al.*, 2024; Woodruff *et al.*, 2023; Yeter *et al.*, 2024; Yue *et al.*, 2024). The remaining eleven do not focus on commercial or market aspects of AI at all.

Studies that explicitly engage with this dimension often describe AI-driven tools as part of a commercialised education ecosystem. Skaar (2024) identifies commercialisation as a key concern, citing issues like data bias, *platformization*, and privacy. Mintz *et al.* (2023) note the prevalence of commercial AI tutors in schools and warn about potential data exploitation and ownership issues.

Some studies highlight economic drivers of AI adoption. Hong & Kim (2024) position AI within industrial transformation, business model innovation, and the digital value chain. Mora-Aristega *et al.* (2023) emphasises how leading AI companies position themselves in the global market, reinforcing AI's growth in education as commercially driven.

Pörn *et al.* (2024) highlight cost and accessibility concerns, warning that unequal AI access could deepen educational inequalities, as well-funded schools integrate advanced AI more easily. Zhao (2022) examines enterprise-school partnerships, weighing benefits and security risks.

Beyond explicit discussions, some studies indirectly reference commercialisation. Du & Lv (2024) highlight the need for AI tools specifically designed for education, suggesting a growing AI education market. They argue that existing generative AI tools lack educational adaptation, reinforcing calls for tailored AI solutions. Some studies recognise private companies' role in AI adoption but do not explicitly link it to commercialisation. Yeter *et al.* (2024) discuss AI-enhanced gamified platforms, reflecting EdTech sector involvement. They suggest that market-driven forces influence AI adoption, positioning schools as consumers of AI-based products, though without broader commercial analysis.

Lucas *et al.* (2024) raise concerns about hidden costs and increased workloads associated with AI. While teachers valued AI for lesson design and homework review, the study does not assess if AI personalisation widens school disparities. Similarly, Pörn *et al.* (2024) mention the high cost of digital tools, such as interactive whiteboards, but do not explicitly link this to AI.

Some studies imply that AI learning platforms are part of a broader commercial landscape. Yue *et al.* (2024) link AI adoption to integration with commercially driven platforms. However, the study overlooks economic drivers of these systems.

Woodruff *et al.* (2023) acknowledge non-profits guiding AI adoption but omit corporate influence. While this underscores institutional involvement, it does not address commercial partnerships or financial interests.

The literature review shows major gaps. While some studies discuss AI's commercialisation, EdTech's role, and data exploitation, few frame these within a broader market analysis. Discussions on costs, economic dependencies, and financial sustainability are largely absent, despite mentions of disparities in AI access. The anticipated risk of market concentration—where a few dominant corporations shape AI education—is only briefly noted, with little analysis of its impact on institutional autonomy. Similarly, while the framework stressed governance challenges in AI-industry collaborations, the literature prioritises technical and ethical concerns over the regulation of commercial partnerships.

A key gap is the lack of scrutiny on AI's long-term financial impact on schools, particularly whether reliance on private providers threatens educational sovereignty. Although the framework warned about AI's influence on curricula and educational priorities, there is little direct evidence of this in the literature. These omissions suggest that economic and market dynamics are overlooked in AI education research, despite their far-reaching implications. Closing these gaps is essential to prevent AI from deepening financial divides, increasing commercial dependencies, or placing corporate interests above educational needs.

4.8. Ideological and belief aspects

Just seven, from the 23 analysed studies, directly discuss ideological and belief aspects (Chen, 2024; Duisenova & Zhorabekova, 2024; Kong & Yang, 2024; Mora Aristega *et al.*, 2023; Pörn *et al.*, 2024; Velandar *et al.*, 2024; Woodruff *et al.*, 2023). Six studies mention ideological aspects indirectly (Lucas *et al.*, 2024; Mintz *et al.*, 2023; Skaar, 2024; Webb *et al.*, 2023; Yeter *et al.*, 2024; Yue *et al.*, 2024). The remaining ten studies did not address issues pertaining to the ideological dimension or belief aspects.

The incorporation of AI into education has significantly influenced underlying beliefs and ideologies related to teaching practices (Kong & Yang, 2024). While some embrace AI as a panacea, others express concerns over its risks, revealing both techno-utopian and technophobic perspectives. This tension underscores the need for a critical approach that carefully weighs AI's potential against its ethical, social, and cultural implications.

The techno-utopian perspective frames AI as a definitive solution to longstanding challenges in education (Duisenova & Zhorabekova, 2024). Much of the current research on AI in education reflects an ideological stance that positions technology as a driver of innovation, progress, and modernization, aligned with global narratives of technological determinism. AI is thus presented not only as an improvement tool, but also as a transformative force capable of reshaping educational experiences. For example, the emphasis on personalized learning reflects beliefs in individual agency and self-direction, which contrast with more traditional, collectivist models that prioritize standardization and group-based learning (Duisenova & Zhorabekova, 2024).

In contrast, the technophobic perspective shows skepticism about AI's accuracy and potential over-reliance on algorithmic solutions in decision-making processes (Skaar, 2024), emphasizes concerns about AI's potential to displace teachers, dehumanize learning, and exacerbate social inequalities (Yue *et al.*, 2024). Technology is perceived as not neutral, but as a threat to the relational, human-centered nature of education, learning process themselves, and even social responsibility and inequality issues (Chen, 2024). Teachers express concern that AI integration may diminish the focus on conceptual learning, mathematical reasoning, and problem-solving, and about AI shifting educational priorities, such as focusing on tools rather than conceptual learning and problem-solving

(Pörn *et al.*, 2024). Additionally, AI is often perceived as accessible only to a select group of students, raising issues of equity and inclusivity (Chen, 2024; Pörn *et al.*, 2024).

Beyond these two polarized positions, the literature also reflects other ideological orientations that can be grouped into at least two additional categories. On the one hand, nuanced critical perspectives are present in studies that neither embrace AI uncritically nor reject it entirely, but instead weigh its benefits and limitations in context. On the other hand, there are naturalized positions, in which the ideological assumptions about technology remain implicit; these studies tend to present AI as a natural or inevitable improvement, without explicitly questioning the broader values and narratives driving its adoption.

Amidst the prevailing discourse on technology, characterised by both technological utopianism and technophobia, critical perspectives emerge to seek a balanced assessment of the opportunities and threats presented by artificial intelligence in education. For example, a study from Lucas *et al.* (2024) revealed that teachers were positive about the benefits of AI-based educational technology but expressed concerns about the perceived lack of transparency. Yue *et al.* (2024) also reflect about teachers' ability to balance optimism and caution when considering the integration of AI in education, highlighting that show both openness to its educational potential and awareness of its limitations. While some express scepticism regarding the accuracy of AI and the risk of over-reliance on algorithmic decision-making, these concerns remain measured and do not reflect technophobic positions. This critical positioning could be related to ethical considerations of the integration of AI in education, as the correlational analysis suggests.

5. Conclusions

Across the reviewed literature—focused on primary education—artificial intelligence is predominantly understood through an instrumental lens. AI is commonly framed as a set of technical tools for enhancing efficiency, personalizing learning, and automating educational processes, often at the expense of deeper pedagogical or critical engagement. AI literacy is typically reduced to operational competence rather than linked to ethical or reflective capacities.

While systematic reviews were excluded from the main coding process due to their meta-analytic nature, their inclusion in the initial corpus offered valuable insights. Two of them (Rizvi *et al.*, 2023; Yim, 2024) addressed all seven dimensions of analysis explicitly, providing a rare example of comprehensive and multidimensional engagement. In contrast, others (Crompton *et al.*, 2024; Sanusi *et al.*, 2023) focused almost exclusively on ethical and instrumental aspects, reinforcing the broader trend of framing AI primarily as a set of tools with moral implications, rather than as sociotechnical systems embedded in power structures. This divergence within reviews themselves highlights how even reflective overviews of the field may reproduce narrow conceptual framings.

Some studies offer a more transformative view, highlighting how AI challenges traditional epistemologies and calls for new frameworks of trust, validation, and inclusive knowledge production. Ethical concerns—particularly around transparency, autonomy, and data privacy—are widely acknowledged, yet often disconnected from broader political and economic contexts. The literature also points to social risks, such as weakened relationships and deepening inequalities, alongside the evolving and sometimes precarious role of teachers.

As AI becomes embedded in educational systems, ethical considerations must guide its implementation. AI offers advantages in personalization, accessibility, and efficiency but raises concerns regarding transparency, autonomy, privacy, fairness, and governance. Establishing an ethical framework ensures AI fosters inclusion, equity, and responsible innovation.

Political, commercial, and ideological dimensions are addressed inconsistently. When they appear, they reveal tensions between innovation discourses and unresolved questions about power, justice, and agency. Overall, the literature tends to treat these dimensions in isolation, limiting the field's ability to grasp the complex and interdependent nature of AI's impact in education.

Taken together, the findings of this review reveal a landscape that is both rich and uneven. While certain dimensions—particularly the instrumental and ethical—are well represented, others remain underdeveloped or only partially addressed. There is a recurring pattern of treating AI as a neutral and inevitable component of educational modernization, often without critically interrogating its broader implications. This tendency to prioritize technical affordances over political, economic, or ideological analysis suggests that current research may be ill-equipped to fully address the complex transformations AI is bringing to primary education.

Furthermore, the review highlights a lack of integration across the different dimensions of analysis. While epistemological considerations occupy a central place in the literature—often closely connected to social and commercial concerns—technical and political aspects remain relatively disconnected, limiting discussions on the governance of AI and development of appropriate education policies. Ethical and ideological dimensions frequently overlap, reflecting unresolved tensions in how AI is conceptualized and framed. Despite the growing commercialization of AI technologies, commercial and social aspects are rarely examined in relation to one another, and political and economic dimensions are often treated in isolation. This fragmented treatment constrains the field's capacity to grasp the complex, interdependent forces shaping AI's development, application, and consequences in primary education. Moreover, the tendency to label any non-technical concern as “ethical”—without engaging with ethical reasoning or frameworks—further illustrates the conceptual shallowness that characterizes much of the literature. This conflation reinforces the need for a multidimensional approach that distinguishes between ethical, social, political, and ideological considerations, rather than collapsing them under a generic ethical umbrella. Advancing theoretical and policy responses will require a more integrative perspective that connects these domains rather than compartmentalizing them.

By mapping how different dimensions are represented—or omitted—this study identifies both patterns and blind spots in current academic discourse. It highlights the need for a more integrated and critical research agenda that moves beyond instrumental rationales. If we are to fully understand the impact of AI on digital literacy in primary education, we must ask not only what AI can do, but also what kinds of knowledge, values, and futures it helps to produce—and for whom.

The uneven treatment of dimensions across the literature—including within systematic reviews—further demonstrates how critical, political, and ideological perspectives remain underdeveloped even in attempts to synthesize the field. This calls for greater reflexivity not only in empirical studies but also in how the research community conceptualizes and consolidates knowledge about AI in education.

This also calls for a broader shift in how responsibility is conceived in the face of AI integration. These questions cannot be left to others. They require active, sustained engagement from all individuals and institutions involved in educational processes—teachers, researchers, policymakers, school leaders, families, and students alike. The choices made around AI adoption are never neutral and must be guided not only by feasibility or innovation, but by a collective and conscious commitment to the educational, social, and ethical purposes of schooling.

At the foundation of any meaningful educational project should lie a commitment to systematic, multidimensional analysis. The seven dimensions explored in this study—instrumental, epistemological, ethical, social, political, commercial, and ideological—must not be treated as peripheral, but as fundamental lenses for understanding and shaping the role of AI in education. Only through this collective responsibility and comprehensive perspective can educational communities ensure that AI serves human development, democratic participation, and educational justice—rather than merely accommodating technological change. In doing so, this study not only addresses the fragmented treatment of AI already noted in critical literature but also reinforces the importance of situating AI research within the broader structures of digital governance, educational privatization, and global EdTech expansion that shape contemporary schooling. This approach challenges researchers, teachers and policymakers alike to advance a conscious, relevant and appropriate integration of AI in primary education from a multidimensional approach to its implications.

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